



Representing geometric sequences graphically

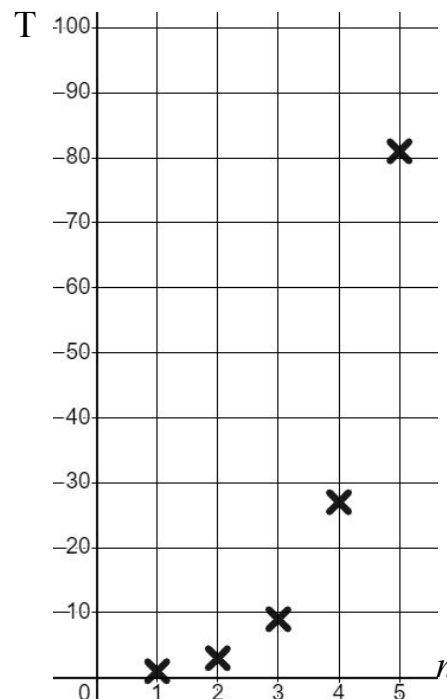
Task A: Graphing a common ratio greater than 1

1) Fill in the table of values and plot the geometric sequence that starts on 1 and has a common ratio of 3

Term number (n)	1	2	3	4	5
Term value (T)	1	3	9	27	81

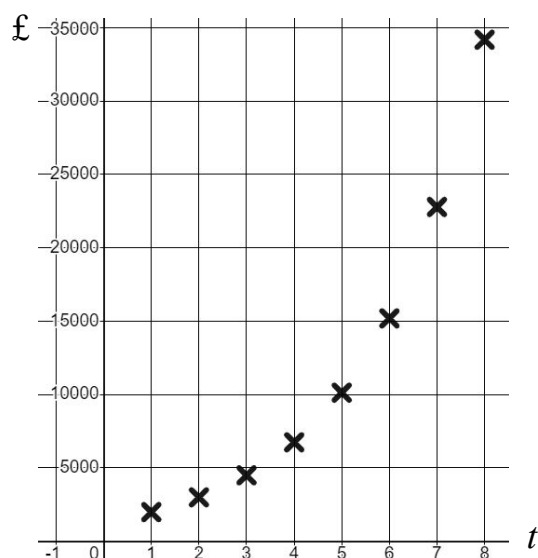
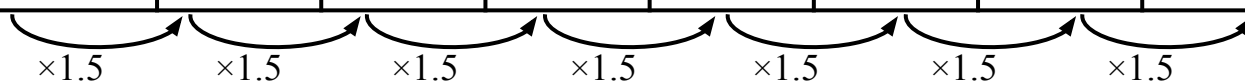


A line helps us to see the curve of a geometric sequence but, we don't draw it.



2) You invest £2000 in a new tech company and your shares grow in value by 50% each year. Calculate their value for the first 8 years and plot it on the graph.

t	1	2	3	4	5	6	7	8
£	2000	3000	4500	6750	10 125	15 188	22 781	34 172



The points plot the curved pattern of a geometric sequence.



3) Sam is contemplating the shape of the graphs of geometric sequences.



I don't know if the starting term affects the shape. If I use the same common ratio of 2 but start on two different numbers will I get a different shape?

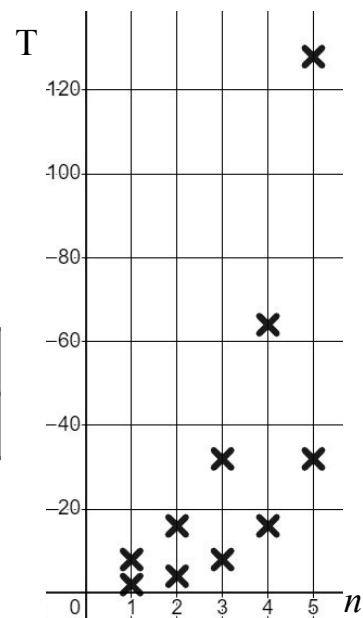
n	1	2	3	4	5
T	2	4	8	16	32

$\times 2$ $\times 2$ $\times 2$ $\times 2$

n	1	2	3	4	5
T	8	16	32	64	128

$\times 2$ $\times 2$ $\times 2$ $\times 2$

Using a common ratio to complete the tables, plot both graphs, and write a conclusion for Sam.



Both graphs have the same shape.

An upward curve getting increasingly more steep.

Task B: Graphing a common ratio less than 1

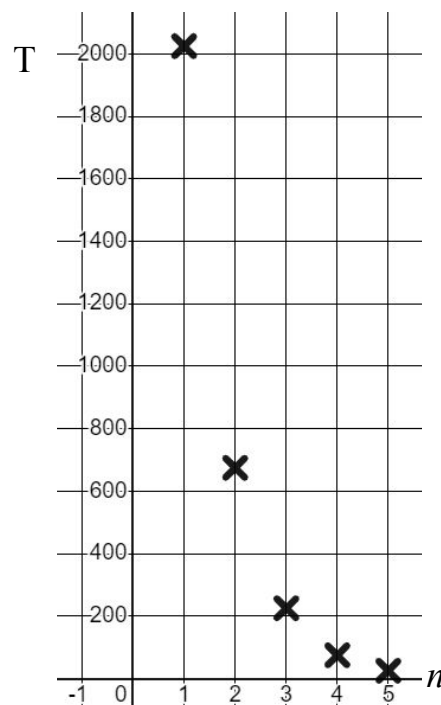
1) a) Complete the table of values and graph this geometric sequence with a common ratio $\frac{1}{3}$

n	1	2	3	4	5
T	2025	675	225	75	25

$\times \frac{1}{3}$ $\times \frac{1}{3}$ $\times \frac{1}{3}$ $\times \frac{1}{3}$

Plot the coordinates as accurately as you can.

Common ratio between 0 and 1 giving us a downward curve.



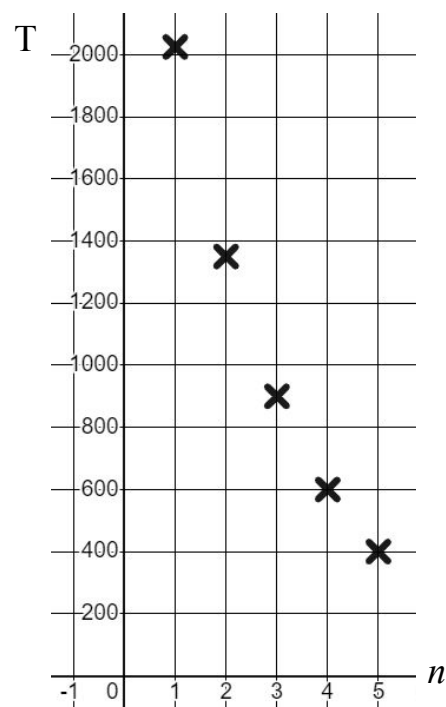
1) b) Complete the table of values and graph this geometric sequence with a common ratio $\frac{2}{3}$

n	1	2	3	4	5
T	2025	1350	900	600	400

$\times \frac{2}{3}$ $\times \frac{2}{3}$ $\times \frac{2}{3}$ $\times \frac{2}{3}$

Plot the coordinates as accurately as you can.

Common ratio between 0 and 1 giving us a downward curve slightly less steep than our $\frac{1}{3}$ curve.



2) Complete the table of values and graph this geometric sequence with a common ratio -1.5

n	1	2	3	4	5	6	7	8
T	2.56	-3.84	5.76	-8.64	12.96	-19.44	29.16	-43.74

$\times -1.5$ $\times -1.5$ $\times -1.5$ $\times -1.5$ $\times -1.5$ $\times -1.5$ $\times -1.5$

Plot the coordinates as accurately as you can.

Negative common ratio giving us oscillating terms with the positive ones forming a curve and the negative ones forming a curve.

